

10 (**short**) Figure 7 shows a radar station tracking an unidentified flying object at a range of 1 km due south moving in a horizontal plane at a speed of $v = 200 \text{ m s}^{-1}$ heading west and with an acceleration $\alpha = 10 \text{ m s}^{-2}$ as shown.

(a) Write down the velocity and acceleration in polar coordinates with the origin located at the radar station, and hence find the radial component of the acceleration. [6]

(b) Calculate the instantaneous radius of curvature of the path. [4]

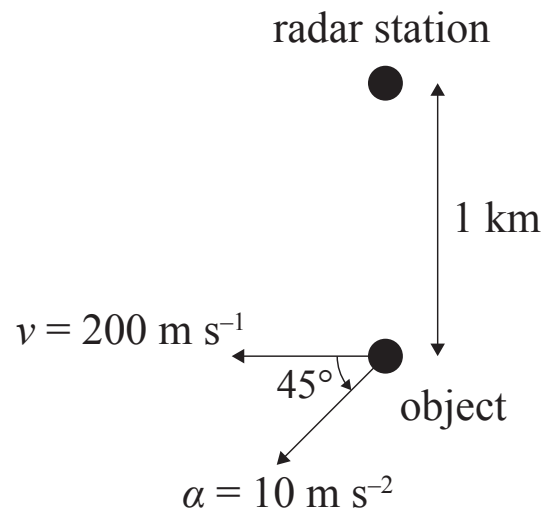


Fig. 7